

8 Read the passage below and answer the questions that follows.

When an earthquake occurs, two different types of seismic waves could be generated: P-waves (primary waves) and S-waves (secondary waves). The speed of the waves depends on wave type and the properties of the rock; the denser the rock, the faster the waves travel.

P-waves travel the fastest. They consist of successive contractions and rarefactions, just like sound waves in air. The motion of the particles in the rocks that the waves travel through is parallel to the direction of the wave. S-waves are slower than P-waves. They are transverse waves, which means that the particle motion is at right angles to the direction of travel. S-waves cannot travel through air or liquids.

By studying the propagation characteristics of seismic waves, scientists have learned much about the detailed nature of Earth's interior. Inner core of Earth consists of solid iron and nickel. Outer core of Earth consists of liquid iron and nickel.

Fig. 8.1 shows the velocity and density variations within Earth (from the surface to the centre) based on seismic observations. The main regions of Earth and important boundaries are labelled.

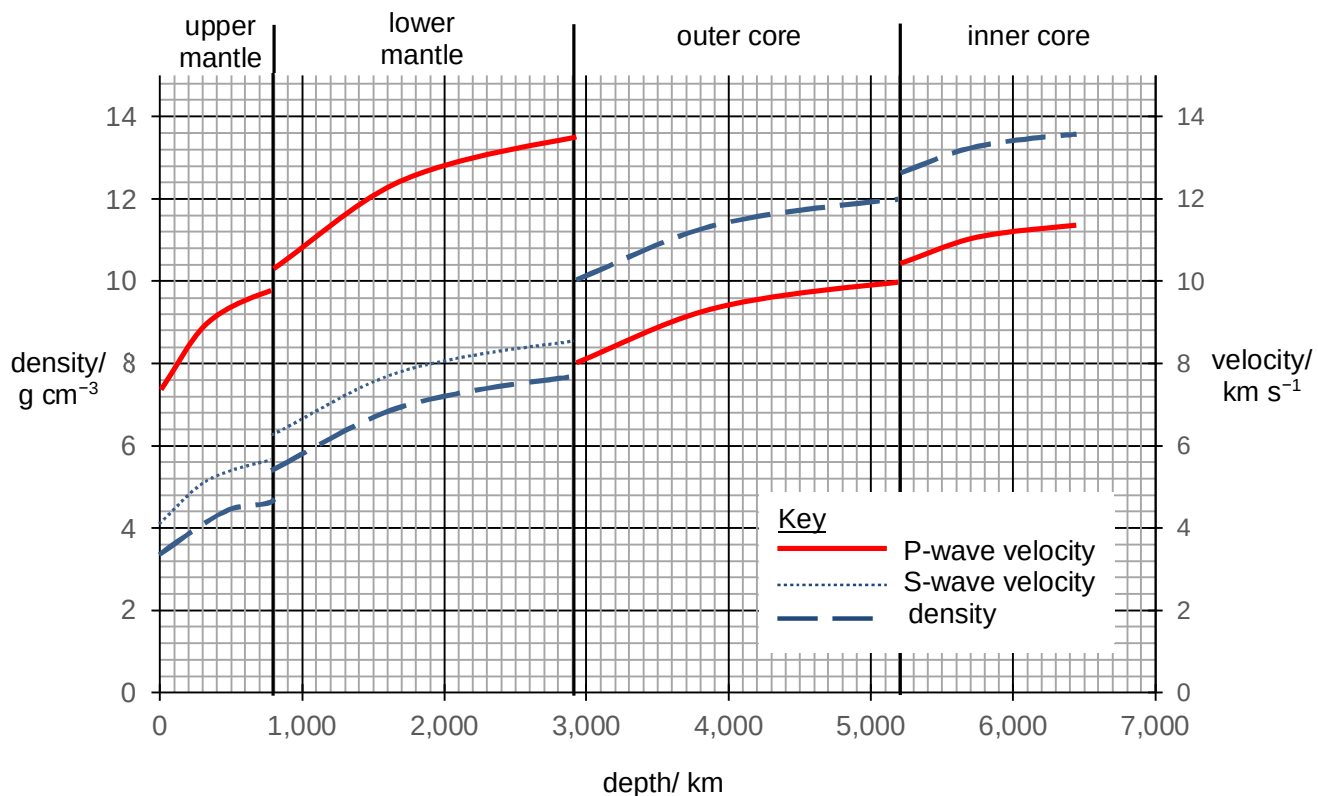


Fig. 8.1

- (a) A large earthquake occurs at a depth of 200 km from the surface of the Earth at a particular location and both seismic P-waves and S-waves are produced. The P-waves and S-waves travel through the Earth away from the epicentre of the earthquake.

A seismograph at a different location detects the arrival of the S-waves 2 minutes 50 seconds after the arrival of the P-waves.

- (i) Estimate the time taken for the S-waves to travel from the epicentre of the earthquake to the seismograph.

time = s [3]

- (ii) Calculate the distance of the epicentre of the earthquake from the seismograph.

distance = km [1]

- (iii) Seismic waves produced by the earthquake are detected by seismographs in many different countries. In certain parts of the world, however, seismographs do not detect any of the S-waves produced by the earthquake.

Using Fig. 8.1 and other information, explain why these seismographs do not detect the S-waves.

.....

 [2]

- (b) The waves from earthquake cause surface vibrations. Surface waves travel along the outer surface of the Earth. These are the slowest and often produce the worst damage. The vibrations can damage buildings far from the epicentre (source) of the earthquake.

Over the past 30 years, it was observed that the vertical component was a likely contributor, along with severe horizontal ground motion, to the mass destructive structural effects observed during extreme earthquakes. It was reported that the magnitude of vertical ground motion exceeded the magnitude of horizontal ground motion especially in strong earthquakes such as Loma prieta 1989, Northridge 1994, Kobe 1995, Kocaeli 1999, and Duzce 1999.

The vertical ground vibrations 50 km from a particular earthquake can be modelled by the equation:

$$s = 0.014 \cos(12t)$$

where s is the vertical displacement in metres and t is the time in seconds.

- (i) State the amplitude and frequency of these vibrations

amplitude = m

frequency = Hz [2]

- (ii) Using differentiation, derive expressions for

1. the vertical velocity v of the ground as these waves pass,

[1]

2. the vertical acceleration a of the ground as these waves pass.

[1]

- (iii) The U.S. Geological Survey measures Peak Ground Acceleration (PGA) to create an instrumental intensity scale for the effects of earthquakes. This scale is shown in Fig. 8.2.

instrumental intensity scale	PGA / m s^{-2}	maximum ground velocity, $v_o / \text{m s}^{-1}$	perceived shaking	potential damage
I	< 0.017	< 0.001	not felt	none
II – III	$0.017 - 0.14$	$0.001 - 0.011$	weak	none
IV	$0.14 - 0.38$	$0.011 - 0.034$	light	none
V	$0.38 - 0.90$	$0.034 - 0.081$	moderate	very light
VI	$0.90 - 1.8$	$0.081 - 0.16$	strong	light
VII	$1.8 - 3.3$	$0.16 - 0.31$	very strong	moderate
VIII	$3.3 - 6.4$	$0.31 - 0.60$	severe	moderate to heavy
IX	$6.4 - 12.2$	$0.60 - 1.16$	violent	heavy
X +	> 12.2	> 1.16	extreme	very heavy

Fig. 8.2

Peak Ground Acceleration (PGA) is the magnitude of maximum acceleration of the ground during an earthquake.

- Assuming that the ground movements can be modelled as simple harmonic oscillations, show that

$$\text{PGA} = 2\pi f \times v_o$$

where f is its frequency and v_o is its maximum ground velocity.

[2]

- Calculate the frequency and the amplitude of the strongest earthquake causing an instrumental intensity scale of V.

frequency = Hz

amplitude = m [3]

3. When a building is shaken by an earthquake, it vibrates at the same frequency as the ground vibration. However, the amplitude of the building's oscillation can be much larger than the amplitude of the movements of the ground.

Explain why the building can oscillate with such a large amplitude.

.....

 [2]

- (iv) It is suggested that the Peak Ground Acceleration is inversely proportional to the distance r from the epicenter.

Fig. 8.3 gives some data from an earthquake.

distance r from epicenter/ km	56	84	120	220
PGA/ m s^{-2}	1.20	0.78	0.53	0.30

Fig. 8.3

Use the data in Fig. 8.3 to explain whether the suggested equation is valid.

[3]